LONGEVITY 19

Urolithin A

Compound made by gut bacteria from ellagitannins in pomegranates, nuts and berries



CLASS
Gut-bacteria metabolite (dietary supplement)

STUDIED FOR
Muscle endurance, mitochondria

HUMAN EVIDENCE
Small human studies

US STATUS
Dietary supplement · not FDA approved

STUDIED FOR

- **Muscle endurance** — HUMAN + ANIMAL
- **Mitochondria** — HUMAN + ANIMAL
- **Running performance** — HUMAN
- **Immune cells** — HUMAN
- **Blood vessels** — HUMAN

Mostly company-funded. Most human trials of urolithin A taken directly were sponsored or funded by its maker. No trial has measured aging or lifespan.

HOW IT WORKS IN THE BODY

Urolithin A ▶ Mitophagy ▶ Worn mitochondria ▶ Muscle function

- **Mitophagy inducer:** helps cells clear out worn mitochondria
- **Extended worm lifespan;** improved exercise capacity in mice and rats
- **Changed mitochondrial gene activity** in human muscle (4 weeks)
- **In people,** effects on aging are not established

EVIDENCE & WHAT'S KNOWN

EVIDENCE

SMALL HUMAN
LEVEL 3 OF 5

- Small randomized trials, 42 to 90 people
- 4 weeks to 4 months long
- Muscle main outcomes mostly not met

NOT YET ESTABLISHED

- Long-term safety
- Half-life in people
- Effect on aging in people

QUICK PRACTICAL INFO

HALF-LIFE
Not established in people

ROUTES STUDIED
By mouth

PHARMACOKINETICS
Blood levels measured in a Phase 1 trial; limited human data

SOLD AS
Dietary supplement

MONITORED IN TRIALS
Walk distance, muscle endurance, CRP, adverse events

SPORT (WADA)
Not on the 2026 list

REGULATORY

FDA	Not approved as a drug
FDA	No questions on GRAS notice 791 for food use (2018)
WADA	Not listed

REFERENCES

1. JAMA Netw Open, urolithin A and muscle endurance in older adults, Jan 2022
2. Nat Med, urolithin A induces mitophagy and extends lifespan in worms, Aug 2016
3. WADA, 2026 Prohibited List

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